

Divay Aggarwal

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Education

Netaji Subhas University of Technology

B.Tech in Electronics & Communication (AI & ML) | CGPA: 8.07

January 2021 – May 2025

New Delhi, India

Summary

Results-driven Software Engineer with expertise in systems development using **C++ algorithms** and **Python scripting**. Proven experience in the **Systems** domain, architecting scalable **Linux foundations** and mission-critical infrastructure software systems processing high-volume diagnostics. Proficient in **Systems Programming**, **Operating Systems**, and **Bash Script automation**. Passionate about maintaining code quality through **Test-Driven Development (TDD)** and applying core OS principles to solve complex concurrency challenges.

Experience

Software Engineer

Reddoorz

December 2025 – Present

Noida, India

- Spearheading the system-level architecture of **"Redbooks"**, a mission-critical financial ledger platform facilitating automated accounting and auditing for over **\$5M+ in monthly transactions**.
- Engineered complex **Python modules** to encapsulate "posting criteria logic," automating the generation of **AR/AP Invoices, Journal Entries, and Credit Memos** with strict **ACID compliance**.
- Designed resilient **ETL pipelines** using **Python automation tools** to synchronize **50,000+ daily records** from mid-office servers to the Redbooks, ensuring 99.9% data consistency.
- Optimized **PostgreSQL** performance via compound indexing and query refactoring, reducing average query execution time by **35%** during peak loads.
- Championed **Test-Driven Development (TDD)** by implementing comprehensive **pytest** test suites, achieving **90% code coverage** and significantly reducing production bugs.

Software Engineer Intern

Reddoorz

August 2025 – November 2025

Noida, India

- Architected a **Smart Budget Management System** for Linux platforms, devising a custom **Weighted Moving Average (WMA)** forecasting formula to predict expected billing with **92% accuracy**.
- Engineered granular **Role-Based Access Control (RBAC)**, defining distinct dashboard views and permissions for **Super Admins, CEOs, and Managers** to oversee budget allotments.
- Implemented **Smart Audit Logs** for compliance and integrated **automated email alerts** via **Python** to notify stakeholders of projected budget overruns, cutting excess spend by 15%.
- Designed interactive **financial visualizations** to display real-time spending trends, optimizing **data structure** aggregations to handle complex reporting data efficiently.

Projects

LinuxFS (Custom Filesystem Monitor) | C++, Python, Linux OS, Bash, ext4

- Architected a systems-level storage module using **C++**, utilizing **Linux kernel interfaces** to generate complex filesystem analytics for virtualized environments.
- Implemented an **automated** debugging tool using **Python** and **Bash**, offloading heavy analysis tasks to reduce system diagnostic times by **40%**.
- Secured subsystems with **Linux** and **ext4** permissions, ensuring strict data isolation and reliability.

SysMonitor (Performance Analysis Tool) | Python, C++, Linux OS, Bash, pytest

- Developed high-concurrency **Linux systems** supporting OS operations, analyzed via **Python** scripts and **Bash**, handling up to 1,000 concurrent processes in load tests.
- Optimized **filesystem** performance using **caching strategies** and enforced data integrity via **C++** algorithms.

Technical Skills

Languages: C++, Ruby, JavaScript (ES6+), Python, Rust, SQL

Backend Frameworks: Ruby on Rails, Node.js, Express.js, Sidekiq

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Tools & DevOps: Git, GitHub, Docker, AWS (S3), Postman, Linux, CI/CD, WebSockets

Core Concepts: Data Structures & Algorithms, REST APIs, MVC Architecture, System Design, OOPs

Achievements

- **Global Rank Top 11%** on LeetCode (Rating: 1714), solving **800+** algorithmic problems across platforms.
- **CodeChef 3-Star Coder** (Rating: 1730), ranked in the top tier of competitive programmers globally.
- Successfully completed the NPTEL **Ethical Hacking** course coordinated by IIT Kharagpur.
- Topped the **Cognitive Psychology** course coordinated by IIT Guwahati, highlighting analytical versatility.